

6.5 Uncertainty in predictive models ↕

Predictive analytics involves using past data to predict future observation values. Techniques and models may be used which can factor in a growth trajectory and seasonality, based on past patterns. A degree of error in the prediction should be expected, as there will be uncertainty in how close a predicted point is to the actual future observation. This is expressed using a prediction interval.

Prediction interval is a range of values within which a future observation is likely to fall (with a certain probability), given what has been observed in the past. Prediction intervals are used in *simple linear regression models*, which are models that approximate the relationship between an *independent variable* and a *dependent variable* with a straight line.

- An independent variable is the variable being used to predict/explain the dependent variable. It is plotted on the “X axis” (horizontal). It can be considered to be the “cause” of the action and can be controlled.
- A dependent variable is the variable being predicted. It is plotted on the “y axis” (vertical). It changes in response to movement in the independent variable and therefore shows the “effect” of the action.

□ Watch: What is predictive analysis? [3.20 mins] and An introduction to linear regression analysis [5.17 mins]

Watch the following two videos, which introduce you to predictive analysis and linear regression analysis. These are useful concepts to understand when creating data visualisations.

As you watch, consider:

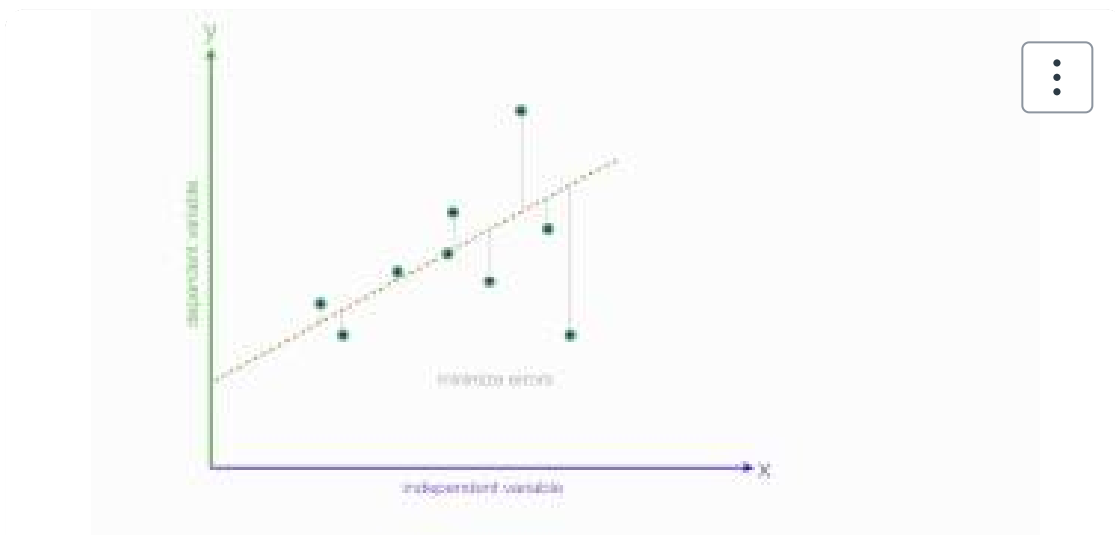
- How does the use of past and current data enable trends to be forecast reliably?
- Why is this important and how it can be used effectively in data visualisations?
- Why is it important to use linear regression analysis carefully in data visualisations?



▶ 🔊 0:00/3:21



Source: © IDG TECHtalk (2020).



▶ 🔊 0:00/5:18



Source: © statisticsfun (2012).

Prediction intervals for time series data

Time series data

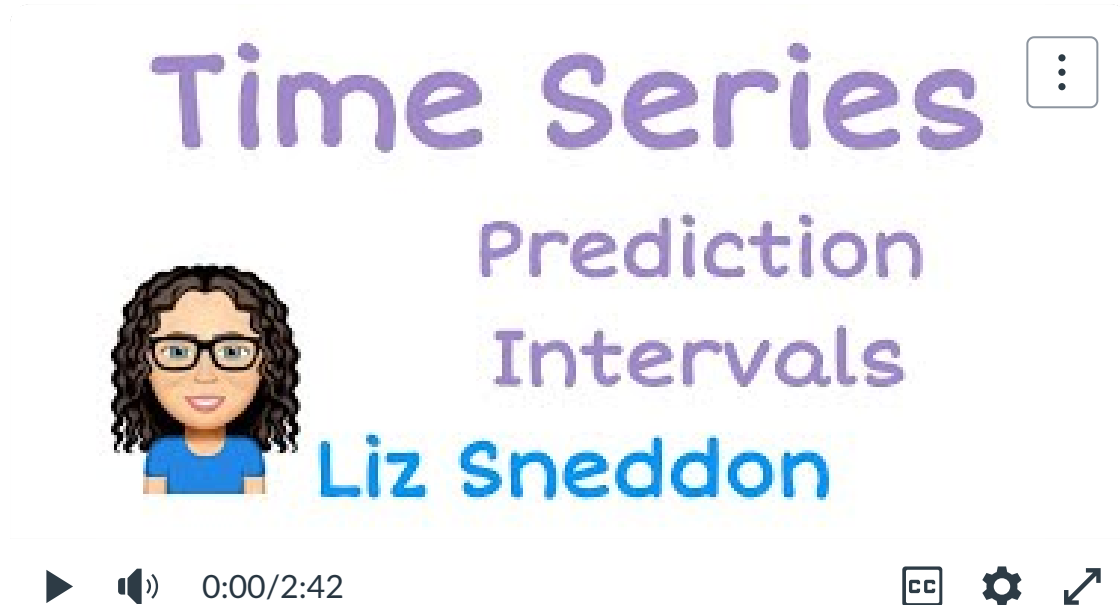
This refers to a series of observations of a variable taken at successive, regular points in time. Observations may be collected every hour, day, week, month, etc. When displaying time, the time unit is displayed along the horizontal axis of a time series chart. The vertical axis is used to display the values of the variable being measured/observed. The temporal nature of the observations is illustrated by a line connecting the consecutive observations.

□ Watch: Time series prediction intervals [2.41 mins]

The following video provides a fundamental understanding about data predictions over time intervals.

As you watch, consider:

- How can the understanding gained from this video be used to improve the data stories that you share?
- Why is understanding prediction intervals important?



Source: © Liz Sneddon (2020).

Note: To maintain a 95% prediction interval, the farther into the future data points are predicted, the wider the prediction interval will become. This is intuitive as the more distant something is, the more difficult it is to confidently predict/understand influencing factors – many things in the future can change.